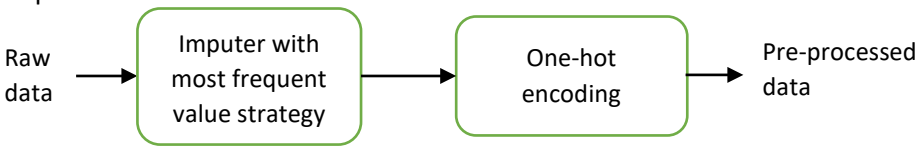
CS714 2024 S1 – Mid-semester test

Question 1: Data preprocessing [4 marks]

Consider the 2 following preprocessing pipelines. For each pipeline, describe likely characteristics of the input raw data and explain how you can derive them from the preprocessing steps.

Pipeline 1

Pipeline 2



(2 marks per pipeline)

For each pipeline, 2 marks are granted if at least one valid insight is derived from each step of the pipeline.

Pipeline 1:

- Input data is numerical with missing values as the pipeline is starting by imputing missing values with the median value of the attribute. Using the median value suggests that the distribution is skewed, as the median is less sensitive to outliers than the mean.
- Raw data might have attributes on different scales as the pipeline is using standardisation.
- Finally, the distribution of some attributes might have a heavy tail as the last step is a log transformation.

Pipeline 2:

- Input data is categorical (text also accepted even if less likely to use on-hot encoding in that case if the vocabulary is large) as one-hot encoding is used. The number of categories is likely not too large as one-hot-encoding would create one new attribute per category. There is likely no ordinal relationship (i.e., natural ordering) between the categories as one-hot-encoding is representing categories as independent from each other.
- There are missing values imputed with the most frequent value, which is commonly used for categorical data.

Question 2: Design choices [6 marks]

You trained a simple neural network with the following settings: 5 hidden layers with ReLU units, 20 neurons per layer, random initialisation, trained for 1000 epochs. The training

accuracy is 95% and the validation accuracy is 60%. For each of the following suggestions, explain if it is likely to improve the validation accuracy and why, if implemented in the next iteration of the training.

- Training for 2000 epochs with the same dataset.
- Training the model on a larger dataset, including additional collected data.
- Using L2 regularisation in each layer.

(2 marks per suggestion - 1 for the answer, 1 for the reasoning)

- Training for 2000 epochs with the same dataset.
  ➔ No, it will not help to increase the accuracy. It will likely worsen the overfitting as the model will learn to fit the training data even more closely.
- Training the model on a larger dataset, including additional collected data.
  ➔ Yes, it might help to increase the validation accuracy if the collected data is relevant to the task and not redundant compared to data already in the dataset. This is because the current model is overfitting and adding more diverse data might help to learn a more general model.
- Using L2 regularisation in each layer.
  ➔ Yes, it might help to increase the validation accuracy. L2 regularisation is likely to help as the current model is overfitting. It will penalise larger weights and encourage to learn a simpler model, less likely to overfit. It does not guarantee an increase in validation accuracy, but it will promote a more generalisable model and reduce overfitting. A more general model is likely to perform better on data not seen on the training, if the bias is low.

Question 3: Evaluation [4 marks]

Consider the following confusion matrix, resulting from the evaluation of a model on a test set containing 1000 instances.

		True class	
		Positive	Negative
Predicted class	Positive	500	400
	Negative	0	100

Calculate the accuracy, precision, and recall. What do you think about this model's performance?

(0.5 per calculation + 2.5 for discussion of performance)

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{500 + 100}{500 + 100 + 400 + 0} = \frac{600}{1000} = 0.60\ (60\%)$$

$$Recall = \frac{TP}{TP + FN} = \frac{500}{500 + 0} = \frac{500}{500} = 1\ (100\%)$$

$$Precision = \frac{TP}{TP + FP} = \frac{500}{500 + 400} = \frac{500}{900} = \frac{5}{9} \approx 0.56\ (56\%)$$

- Observation: The model is very good at classifying true examples, but it does poorly at classifying negative examples. Therefore, the overall accuracy and the precision are not very good, and the recall is very high. 1.5 marks
- Explanation: The model seems to be tuned to be very sensitive (i.e., positive prediction threshold very low) and ends up predicting most examples as positive. It looks like it is doing a good job at distinguishing positive examples, while it is actually predicting almost everything as positive. (1 mark)

Question 4: Learning rate and optimisers [4 marks]

1) Give an example of a learning rate schedule and explain briefly why it might be beneficial when training a Deep Neural Network (DNN).
2) Explain in your own words what is the momentum mechanism and what is its effect on the gradient descent optimisation process.

1) 1 possible example: Exponential lr decay. Reducing the lr as the training goes might be beneficial to go quickly towards an optimum at the start and slowing down to avoid overshooting it. (2 marks)
2) The momentum mechanism keeps track of the direction of past gradients (with an exponentially decaying sum) to adjust the direction of the gradient descent steps. This allows to take steps in a direction closest to the true gradient and helps speed up learning. (2 marks)

Question 5: RNN and Transformer [4 marks]

The RNN architecture processes each input in a sequence sequentially.

1) Explain why this is at the same time an advantage and a drawback of this architecture.
2) Explain how the Transformer architecture processes inputs (on the encoder side) to alleviate this drawback of RNNs.

1) This is an advantage of RNNs because they make the implicit assumption that the data is ordered, and this information is capture "naturally" during training. (1 mark) It is also a drawback because it leads to long processing time for long sequences. (1 mark)
2) The Transformer architecture uses embeddings to create low dimensional representation of the input sequence, instead of processing it sequentially. This speeds up the processing time, but the information about the sequence order is lost. (1 mark) To fix this, the Transformer uses positional encoding to integrate information about the sequence order in the embedding. (1 mark)

Question 6: CNN feature map [4 marks]

Answer the following questions about feature map dimensions on CNN.

1) What will be the dimensions of the feature map at the output of a convolutional layer using ten $5 \times 5 \times 5$ filters, a stride of 3 and a padding of 0, knowing that the input dimension is $50 \times 50 \times 5$? Show your calculation steps.
2) What will be the dimensions of the feature map at the output of an average pooling layer using a 5×5 filter and a stride of 5, knowing that the input dimension is $50 \times 50 \times 5$? Show your calculation steps.
3) Same question than 2), this time using a max pooling layer.

1) $((n+2\times p-f) / s) + 1 = ((50+2\times 0-5)/3) + 1 = 45/3 + 1 = 15 + 1 = 16$
   Feature map dimensions: $16 \times 16 \times 10$ (10 = number of filters) (1.5 marks)
2) $((n-f)/s) + 1 = ((50-5)/5) + 1 = 9 + 1 = 10$
   Feature map dimensions: $10 \times 10 \times 5$ (5 = depth of input) (1.5 marks)
3) Same answer as 2), max or average does not change the output dimensions, it would only change the values in the feature map. (1 marks)

Question 7: DNN training [4 marks]

1) Explain why using neural networks with a lot of layers and weights per layer make the optimisation process harder.
2) Give two possible strategies which make the training of deep neural networks easier, and explain briefly their effect on the training.

1) More prone to vanishing/exploding gradients / more likely to overfit / longer training time. (2 marks if at least 2 valid reasons are given)
2) Several possible choices: regularisation against overfitting, using specific architectures (e.g., LSTM, GRU), using tricks like skip connections or auxiliary classification branches (e.g., in GoogLeNet or ResNet), using batch norm, using different optimisers (e.g, RMSProp, Adam), etc.. (1 mark for each strategy)